

Kanokpon Mettasat

Electrical & Computer Engineering, Institute Honors Program -- post-quantum cryptography & blockchain security research

mettasatk27@vmi.edu | Lexington, VA | github.com/gun321h-creator | linkedin.com/in/kanokpon-mettasat | ORCID 0009-0002-6260-8196

ABOUT

ECE cadet at Virginia Military Institute (Class of 2027, Institute Honors Program), applying to MS programs in Electrical Engineering for Fall 2027: secure and embedded systems, post-quantum cryptographic engineering for resource-constrained devices. Undergraduate researcher on quantum-resistant blockchain security (first-author paper accepted at IEEE UEMCON 2026; second under revision at Computers, MDPI). Hardware lead on a fabricated KiCad PCB; team lead at IEEE SoutheastCon 2026 (top seven of forty); sole developer of Officer Parker, a real-time voice agent. President, Eta Kappa Nu (IEEE-HKN), VMI, 2026-27.

EDUCATION

Virginia Military Institute -- BS, Electrical and Computer Engineering (Aug 2023 -- May 2027 (expected))

- Institute Honors Program; Dean's List; Overall GPA 3.81 / Major GPA 4.00.
- President, Eta Kappa Nu (IEEE-HKN), VMI chapter (2026 -- 2027); Tau Beta Pi, Virginia Chapter (Nov 2025).
- IEEE SoutheastCon 2026 -- top 7 of 40 teams, Undergraduate Circuit Design (Mar 2026).
- Chulachomklao Royal Military Academy (CRMA), Class 74 -- selected first of the class for the Royal Thai Army's overseas study programme to the United States (2022).

PUBLICATIONS

- K. Mettasat and E. Fathalla. "Post-Quantum Cryptography for Blockchain: A Systematic Mapping by Cryptographic Family." *2026 IEEE 17th Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON), Armonk, New York*, Oct 2026. Accepted; to appear in IEEE Xplore.
- K. Mettasat, E. Fathalla and M. Azab. "A Systematic Mapping of Quantum Threats and Defenses for Blockchain." *Computers (MDPI), Special Issue on Intelligent Systems Security*, 2026. Under revision following first-round review.

PROJECTS

Officer Parker -- real-time voice agent for mock US visa interviews.

Browser voice agent: 24 kHz PCM capture, speech-to-speech WebSocket session, structured report from the transcript, grounded in the applicant's own DS-160 so contradictions surface. Sole developer; AssemblyAI Voice Agent Hackathon (Sep 2026); MIT-licensed, in production.

JavaScript

Web Audio (AudioWorklet, 24 kHz PCM)

WebSocket speech-to-speech

Vercel serverless functions

Statistical research platform for financial time series.

Python research stack bound to acceptance thresholds registered before any experiment ran (bootstrapped lower-bound Sharpe, deflated Sharpe, multiple-testing correction); 60+ hypotheses rejected by the data; mutation-tested guards.

Python

pandas / NumPy

Event-driven simulator

Block bootstrap

SURI -- Systematic mapping of quantum-resistant blockchain security.

Undergraduate research with Dr. Efath Fathalla (VMI ECE): literature screened against a pre-registered protocol with a 76+ query search log; idempotent Python pipeline rebuilds the corpus from the log. Paper accepted at IEEE UEMCON 2026; second under revision at Computers (MDPI).

IEEE Xplore

Pre-registered protocol (PRISMA-style)

Multi-dimensional taxonomy

Coverage matrix

SKILLS

Languages: Python

Hardware: KiCad (schematic to fabricated PCB)

Engineering: Digital systems design

Research: Systematic review methodology

Tooling: Git

Web / Infrastructure: FastAPI

Live portfolio: km-portfolio.onrender.com/portfolio